



Society of Petroleum Engineers

SPE-227738-MS

Micro-Fracturing in Depleted and High Permeability Reservoir

G. M. Hashmi, Halliburton, Houston, Texas, USA; M. I. Rahim, Petronas, Kuala Lumpur, Malaysia; S. Han, SLB, Kuala Lumpur, Malaysia; M Abdein, Halliburton, Kuala Lumpur, Malaysia; O. L. Wah and N. A. B. M. Dahlan, Petronas, Kuala Lumpur, Malaysia; C. E. Vazquez, Halliburton, Kuala Lumpur, Malaysia; L. S. Ting, SLB, Kuala Lumpur, Malaysia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227738-MS](https://doi.org/10.2118/227738-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Understanding in-situ stress conditions is essential for the safe and efficient drilling and completion of wells, particularly in depleted and high-permeability reservoirs where wellbore stability becomes a significant concern. This paper presents a field case study involving the successful application of wireline-conveyed micro-fracturing in such a challenging environment. The primary objective was to accurately determine breakdown pressures and minimum principal stress to calibrate and refine geomechanical models used for well planning. The outcomes of this work directly contributed to the optimization of drilling parameters and led to substantial cost savings in the development of subsequent wells.

Micro-fracturing on wireline comprises of straddling the zone of interest between packers and creating small-scale hydraulic fractures to obtain information about stresses in the reservoir rock. Once the injection stops, the built-up pressure dissipates into the formation causing the fracture to close, yielding the minimum principal stress. Micro-fracturing operations were performed using a formation testing tool string equipped to isolate four target reservoir zones between inflatable packers. Small-scale hydraulic fractures were induced, and real-time pressure data was monitored for breakdown and closure signatures. The pressure fall-off after injection cessation was analyzed using G-function and square-root time methods to determine minimum principal stress. The paper details the design and execution of these operations, emphasizing the integration of reservoir-specific properties and geomechanical understanding into test planning.

Of the four zones tested, three yielded successful fracture events, while one did not experience breakdown within the tool's operational limits. Nonetheless, even the unsuccessful attempt provided valuable lower-bound estimates for fracturing pressure. The minimum principal stress measured in two of the zones aligned well with pre-test model predictions, while the other two showed significant deviation. These discrepancies underscored the importance of direct measurements in reducing model uncertainties. Based on the findings, geomechanical models were recalibrated and applied to future wells in the field.

The study demonstrates the technical viability and advantages of micro-fracturing in depleted, high-permeability formations. Compared to conventional stress estimation methods, this approach allows for precise fracture placement, lower fluid volumes, and reduced environmental footprint. Importantly, the

recalibrated models informed decisions that led to reduced reliance on Managed Pressure Drilling (MPD) techniques. As a result, the operator avoided significant operational complexities and realized an estimated cost saving of approximately USD 4.5 million.

While micro-fracturing is an established technique, its deployment in depleted reservoirs has been limited due to concerns over well control and formation damage. This study highlights a successful execution under such conditions, providing valuable insights into risk mitigation, model enhancement, and cost-effective well construction. The outcomes offer meaningful guidance for engineers and decision-makers working in similarly challenging reservoir environments.

Introduction

As oilfields mature, reservoir depletion becomes an inevitable consequence of prolonged production. Depletion alters the subsurface stress regime, typically resulting in decreased pore pressure and modifications in horizontal stress magnitudes. In high-permeability formations, these changes are further complicated by rapid pressure dissipation, making wellbore stability a moving target.

Geomechanics offers a framework for predicting subsurface mechanical behavior, and its application in drilling engineering has gained traction over the last two decades. By integrating well logs, core data, and pressure measurements, geomechanical models can predict key parameters such as pore pressure, stress magnitudes, rock strength, and wellbore failure potential (Zoback, 2007). However, such models are heavily reliant on the quality and quantity of input data, which is often limited in depleted reservoirs due to economic and operational constraints (Fjaer et al., 2008).

To overcome data scarcity, this study initiated a comprehensive geomechanical modeling campaign supported by direct in-situ stress measurements via micro-fracturing. This technique offered a practical means to validate and calibrate stress models (Warpinski et al., 1985), particularly the estimation of minimum horizontal stress – a crucial input for mud weight design and fracture gradient prediction.

Accurate estimation of the minimum in-situ stress is fundamental to numerous applications in the oil and gas industry. Key domains where such knowledge is indispensable include wellbore design, drilling and completion analysis, reservoir stimulation, and caprock integrity assessment. Among the principal stress components, the minimum horizontal stress plays a particularly critical role throughout the entire life cycle of a well, from planning through to production and abandonment.

In the context of well design and geomechanical modeling, understanding the spatial variation of minimum horizontal stress across the stratigraphic column is essential for predicting wellbore stability. This stress component, in conjunction with borehole collapse pressure and the effective fracture gradient, defines the safe mud weight window – arguably the most influential parameter in drilling geomechanics aimed at mitigating instability risks. Inadequate characterization of this window can lead to costly non-productive time (NPT) due to issues such as lost circulation or wellbore collapse.

In hydraulic fracturing operations, the stress contrast between the target reservoir and the bounding formations governs fracture height growth and geometry (Barree and Miskimins 2015; Grieser et al. 2007; Ishijima and Roegiers 1983; Lecampion et al. 2013). Accurate profiling of the minimum horizontal stress is therefore vital for effective fracture containment and optimization. It directly influences the selection of perforation intervals and stimulation strategies, thereby enhancing reservoir contact while minimizing the risk of out-of-zone growth.

Similarly, in caprock integrity evaluations, knowledge of the minimum horizontal stress is indispensable. The differential stress between the reservoir and the overlying seal formation determines the critical threshold pressures for various injection processes. These include water injection, steam-assisted gravity drainage (SAGD), CO₂ sequestration, cuttings reinjection, and gas storage operations. Misestimation of this parameter may compromise containment and lead to potential environmental and operational hazards.

To reduce uncertainty across these applications, it is essential to calibrate the stress profile using direct or indirect measurements at multiple scales. At the smallest scale, microfracture testing involves the injection of a small fluid volume into a ~1-meter open-hole interval isolated between straddle packers to induce a tensile fracture and evaluate closure pressure. At a larger scale, extended leak-off tests (XLOTs) conducted below the casing or liner shoe inject greater fluid volumes into 3-5 meter open-hole intervals to initiate and propagate fractures, thereby providing bulk-scale estimates of the minimum horizontal stress. Integration of such multi-scale data into geomechanical models significantly enhances the reliability of stress profiling for field development planning.

This paper documents the geomechanical modeling process, planning and execution of the micro-fracturing operations, and the integration of results into updated stress models. The impact of these efforts on subsequent drilling campaigns is also discussed.

Micro-fracturing

The use of micro-fracturing on wireline to conduct downhole Diagnostic Fracture Injection Tests (DFITs) represents a significant advancement in reservoir characterization techniques, particularly in unconventional and tight formations. This method enables precise, controlled initiation of small fractures using a wireline-conveyed tool that eliminates the need for coiled tubing or conventional hydraulic fracturing spreads. By isolating the interval and pressurizing it incrementally, micro-fracturing allows for the collection of high-resolution pressure falloff data, which can be used to determine key formation properties such as minimum in-situ stress, formation permeability, and leak-off characteristics. The wireline deployment greatly reduces operational complexity and cost, while enhancing the ability to perform tests at multiple depths during a single run. Moreover, this approach minimizes formation damage and fluid invasion due to the limited injection volumes, making it ideal for tight formations where fluid sensitivity is a concern. In high permeability or depleted reservoirs, wireline micro-fracturing allows for greater control over fracture size and avoids excessive fluid loss, ensuring more accurate geomechanical modeling. This wireline methodology has been successfully implemented in permeable reservoir sands, as demonstrated by [Bautmans et al. \(2013\)](#), where fracture closure was primarily governed by fluid losses into the formation, enabling the application of conventional pressure fall-off analysis techniques. Additionally, this approach has been extended to low-permeability formations, with results reported by [Mishra et al. \(2011\)](#) and [Malik et al. \(2015\)](#). [Ardila et al. \(2009\)](#) further contributed to the literature by presenting outcomes obtained using a wireline-conveyed cased hole testing system. The integration of microfracture-based DFITs into early-stage reservoir evaluation supports optimized stimulation design and improved reservoir management decisions. [Fig. 1](#) shows the typical bottomhole interval pressure profile that is observed during micro-fracturing operations.

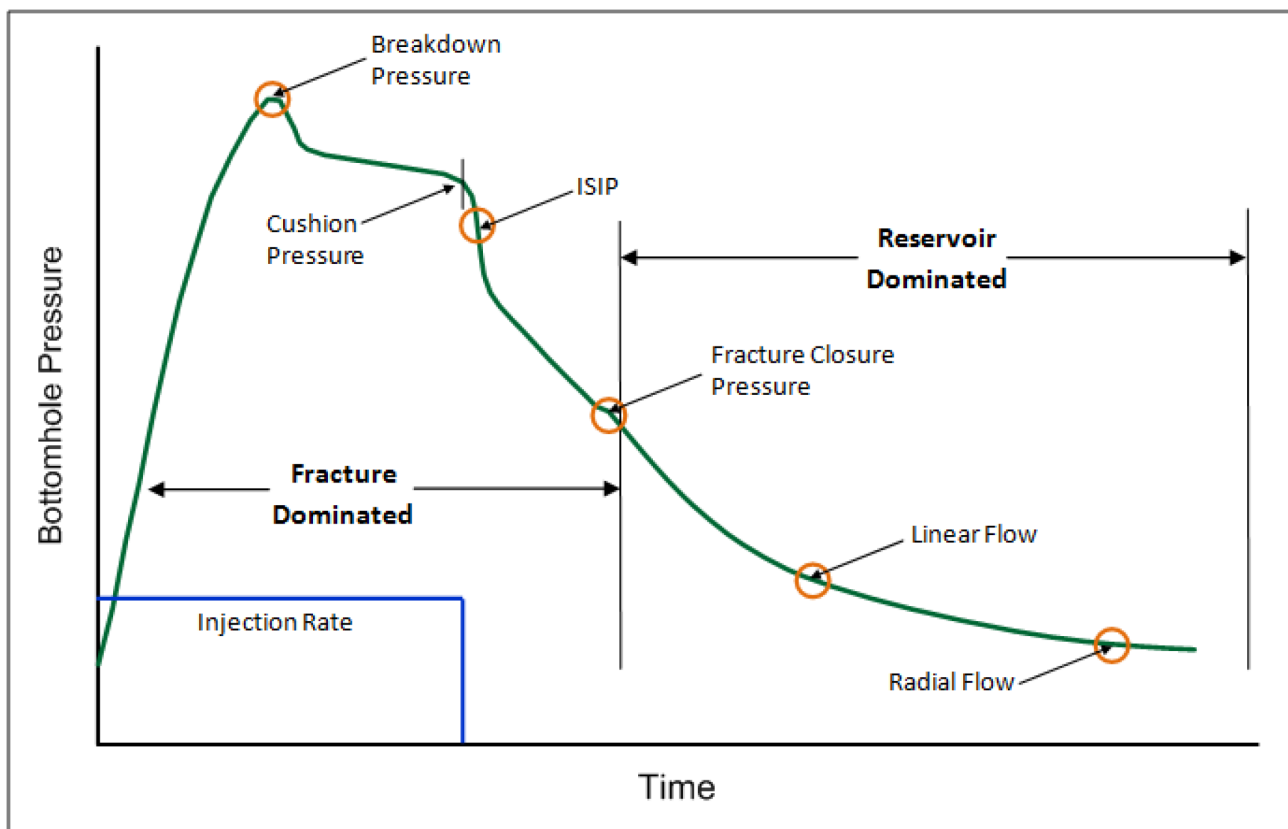


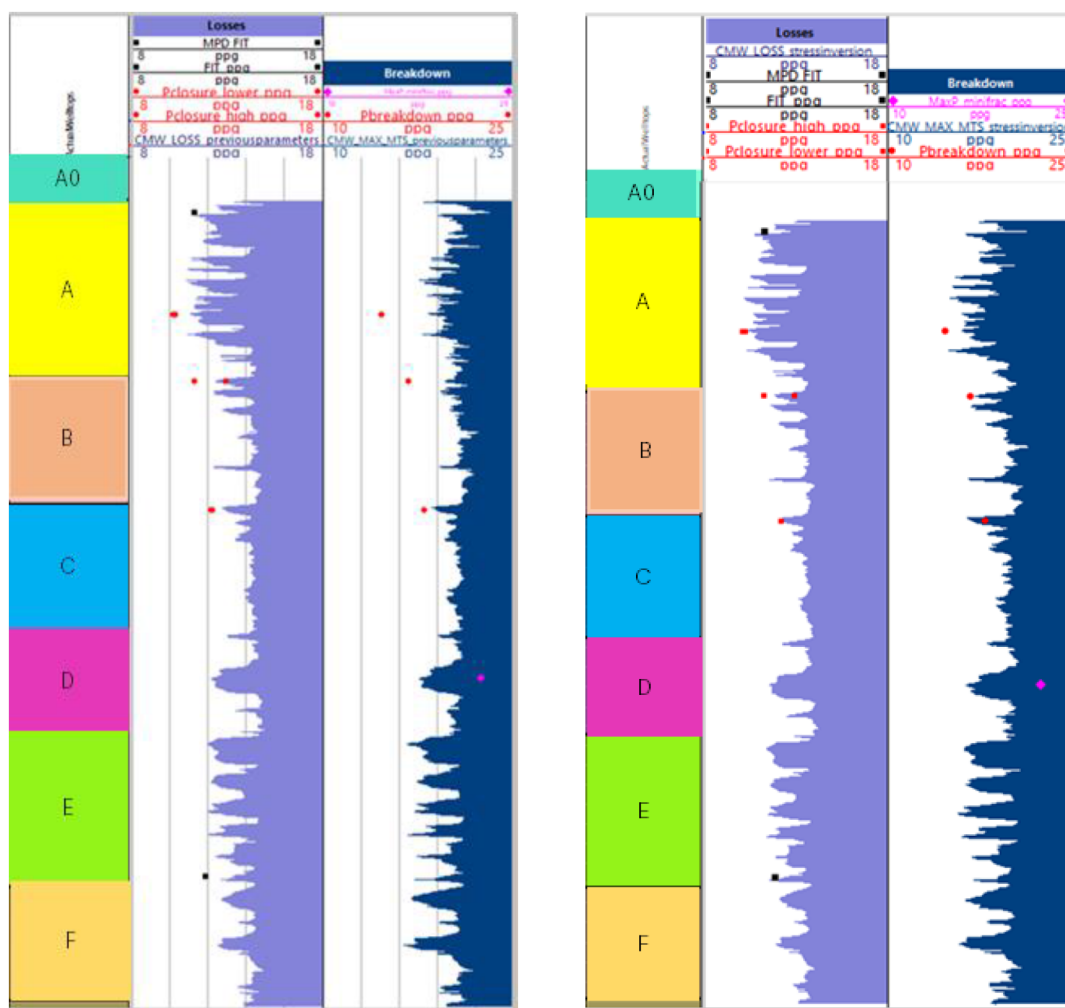
Figure 1—Micro-fracturing Typical Pressure Profile

Planning

Planning such operations in a depleted and high-permeability environment requires careful consideration of both geological and operational constraints. The planning phase began with a thorough analysis of offset well data, including formation pressures, rock properties, and prior leak-off tests. This information was used to construct an initial geomechanical model, which predicted the stress state and informed the design of the microfracture test program. The initial geomechanical model is shown in Fig. 2 on the left with the calibrated model shown on the right. Key goals of planning included:

- Identifying suitable depth intervals based on lithology and permeability.
- Estimating breakdown and closure pressures.
- Selecting appropriate tools and packers based on expected pressure and flow conditions.
- Ensuring test intervals could be isolated effectively despite formation heterogeneity.

The chosen toolstring incorporated straddle packers, pressure and temperature gauges, and a downhole flush pump. Zone selection targeted four distinct sand intervals – Sands A, B, C, and D – each representing different reservoir characteristics.



Planning also involved risk assessment, particularly regarding tool capacity, potential fluid losses, and packer inflation pressures. Special attention was given to the tool's functionality in high permeability environments to determine rates in excess of the cumulative leak-off rate into the formation. Simulations were conducted to determine injection rates that would initiate fractures while maintaining operational safety margins. Final test design recommended injection rates between 15 to 35 ccps and test durations of up to 30 minutes per cycle.

General Procedure

Microfracturing tests were conducted using a straddle packer system deployed at one-meter intervals. The toolstring was equipped with downhole pressure and temperature gauges to ensure high-fidelity data acquisition, along with a downhole flush pump for controlled fluid injection. As illustrated in Fig. 1, the straddle packer was accurately positioned across the target interval within each sand unit. Fluid was then pumped into the packer elements to inflate them, thereby establishing a hydraulic seal and isolating the test interval from the surrounding borehole. The required inflation volume and time were dependent on the borehole geometry and the injection rate, with inflation rates maintained between 25 to 30 ccps. Upon successful inflation and confirmation of the packer seal, fluid injection into the formation commenced.

Bottomhole pressure increased progressively until it reached the formation breakdown pressure (FBP), which was identified by a characteristic and abrupt pressure drop, signifying the initiation of fracture

propagation. This was followed by a relatively stable pressure phase, denoted as the Fracture Propagation Pressure (FPP). After cessation of injection, pressure declined in a manner governed by the formation's permeability; higher permeability formations exhibited a more rapid pressure falloff. During this falloff period, pressure data were analyzed to identify the Instantaneous Shut-In Pressure (ISIP), representing the pressure required to maintain an open fracture, and the Fracture Closure Pressure (FCP), which is critical for determining the minimum horizontal stress (S_{h_min}).

Multiple injection and falloff cycles were performed at each microfracture test station to enhance the reliability of minimum stress estimations, reduce the influence of near-wellbore effects, and confirm fracture re-opening pressure. Pre-job simulations and planning facilitated the selection of optimal target intervals within each zone. Of the four zones tested, three yielded high-quality fracture data. Figures 3 through 6 present the pressure profiles recorded during straddle packer inflation and deflation across all four test stations, including comparative results from multiple injection-falloff cycles for the A, B, C, and D sand intervals. The following sections illustrate the observations per zone:

Station # 1 in Sand A. The test was carried out in three injection cycles, with successful formation breakdown achieved during the first cycle. An injection rate of 15 ccps was maintained in Cycle 1, and subsequently increased to 25 ccps for Cycles 2 and 3 to enhance fracture propagation. The cumulative injection volume across all three cycles was approximately 43 liters, with a total injection duration of around 101 minutes. Consistent and repeatable measurements were obtained across all cycles, indicating high data reliability and stable test conditions.

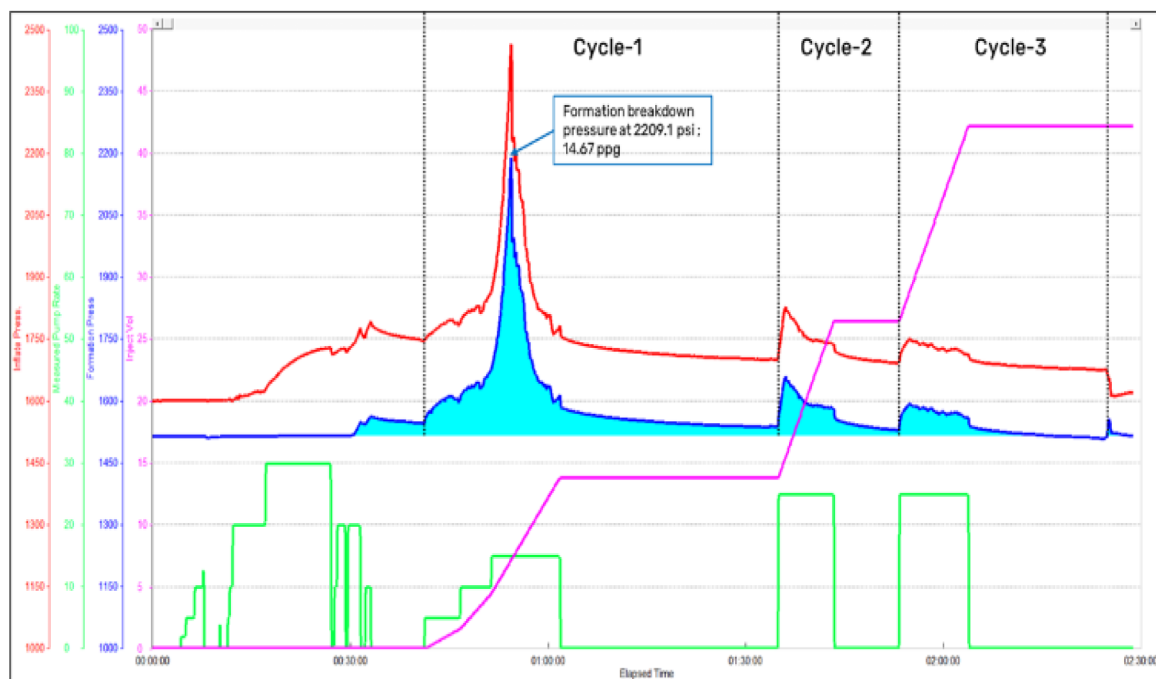


Figure 3—Multiple Cycles of Injection and Fall-off Tests - Station # 1 in Sand A

Station # 2 in Sand B. The test was performed over four injection cycles, with successful formation breakdown achieved on the first attempt. A constant injection rate of 25 ccps was maintained throughout all four cycles. The total cumulative injection volume and duration were approximately 85 liters and 175 minutes, respectively. Notably, the fracture propagation pressure observed in Cycle 1 was relatively higher, likely due to a shorter injection duration. However, the subsequent three cycles demonstrated good repeatability in pressure response, indicating stable fracture behavior and reliable data acquisition for geomechanical analysis.

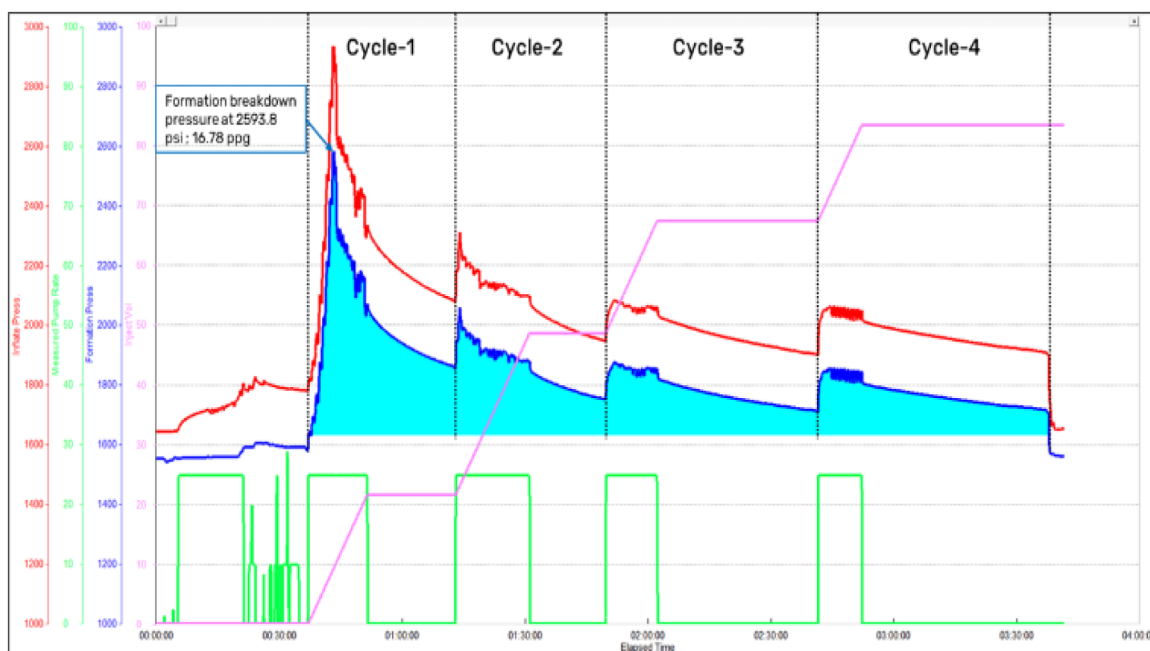


Figure 4—Multiple Cycles of Injection and Fall-off Tests - Station # 2 in Sand B

Station # 3 in Sand D. In the D Sand interval, multiple injection cycles were conducted; however, the formation could not be successfully broken down. The test was ultimately constrained by the maximum pressure rating of the straddle packer system, which was reached during the attempts.

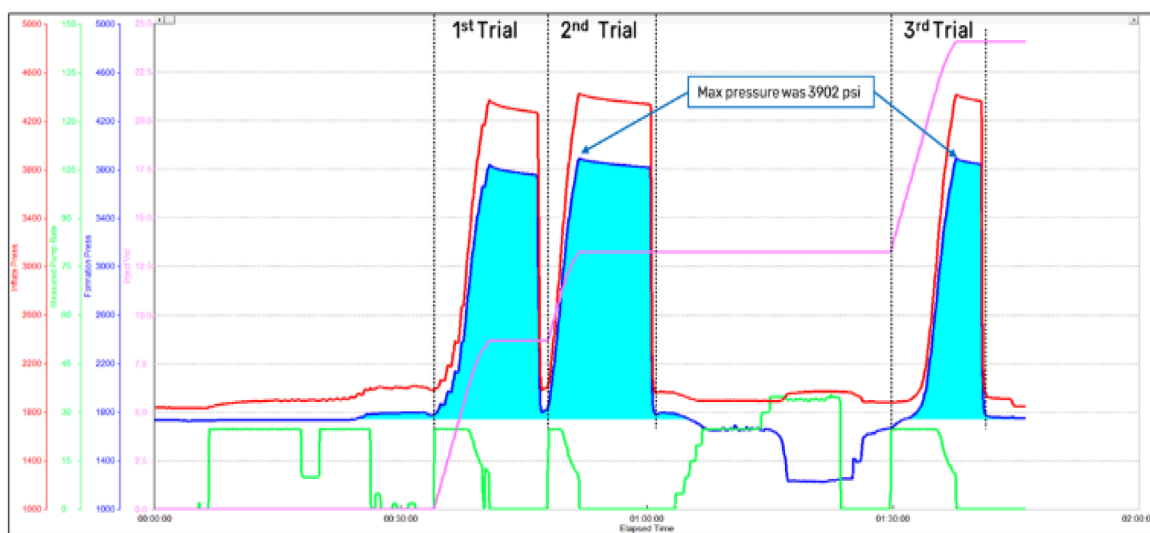


Figure 5—Multiple Cycles of Injection and Fall-off Tests - Station # 3 in Sand D

Station # 4 in Sand C. The test was completed over three injection cycles, with successful formation breakdown achieved during the initial attempt. An injection rate of 25 ccps was maintained for the first two cycles, and increased to 35 ccps in the third cycle to enhance fracture propagation. The total cumulative injection volume and duration across all three cycles were approximately 79 liters and 114 minutes, respectively. The pressure fall-off data from Cycle 1 exhibited considerable noise, leading to notable variability in the measurements compared to the more consistent and reliable results obtained in Cycles 2 and 3.

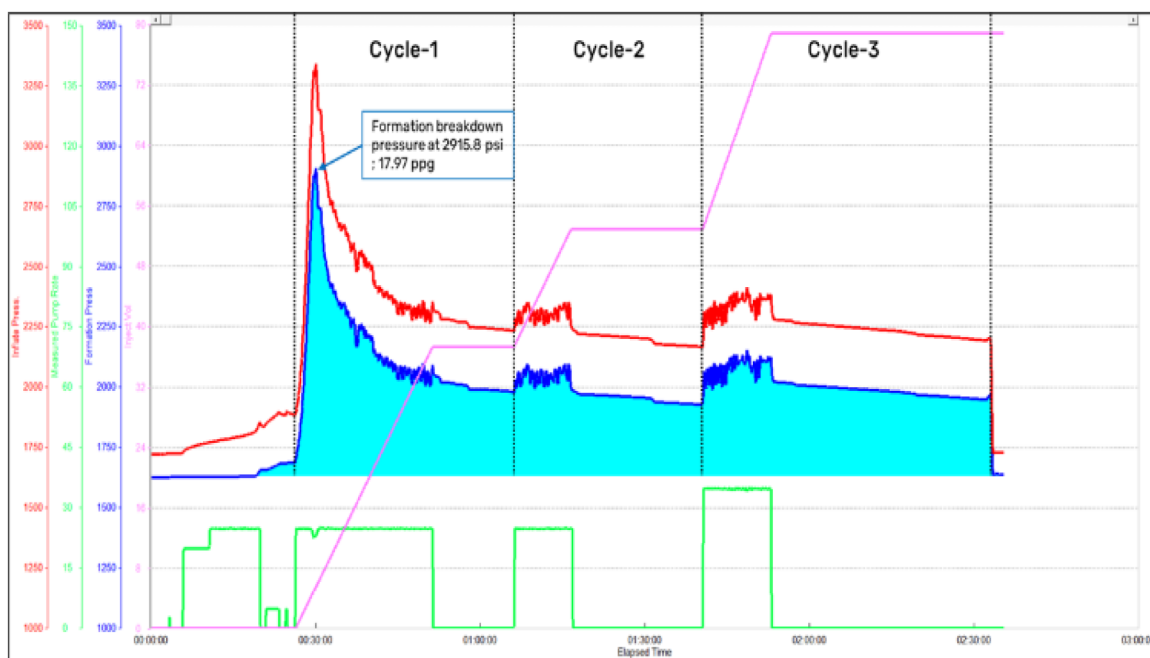


Figure 6—Multiple Cycles of Injection and Fall-off Tests - Station # 4 in Sand C

The pressure-time plots showed the expected behavior for micro-fracturing in permeable formations – rapid pressure drop post-injection due to leak-off, followed by stabilized closure.

Closure Pressure Analysis

Analyzing micro-fracturing data for closure pressure is a critical component of Diagnostic Fracture Injection Tests (DFITs), as it provides valuable insights into the in-situ stress regime of the reservoir. Closure pressure is typically interpreted as the minimum horizontal stress and serves as a key input for geomechanical modeling and fracture design. One of the most commonly used techniques for identifying closure pressure is the G-function analysis, which normalizes time and pressure data to account for leak-off behavior in the formation. By plotting pressure decline against the G-function or its derivative, analysts can identify the point at which the pressure decline deviates from linearity, indicating fracture closure. This method is particularly effective in formations with moderate to high leak-off, where pressure responses are more distinguishable.

Another widely adopted approach is the square root of time analysis, which involves plotting pressure data against the square root of shut-in time. In this method, the linear portion of the plot represents pressure decline dominated by fluid leak-off, with the inflection point on the plot interpreted as fracture closure. This technique is simpler to implement and can be useful when G-function plots are inconclusive or when data quality is limited. However, it is more susceptible to interpretation errors in formations with complex leak-off behavior. Advanced workflows often incorporate both methods, along with specialized diagnostic plots such as log-log derivatives and compliance analysis, to cross-validate closure pressure estimates. By combining these techniques, engineers can enhance the robustness of their interpretations and reduce uncertainty in geomechanical models, ultimately supporting more efficient and effective stimulation strategies in both tight and high-permeability reservoirs.

Table 1 summarizes the results from all stations where successful formation breakdown was achieved. Figures 7 through 19 present the detailed analysis plots for each injection cycle at the respective stations. In instances where individual cycles exhibited noisy data, making it challenging to accurately determine a unique closure pressure, the presence of multiple cycles at each station helped mitigate this uncertainty and supported more robust interpretation of the results.

Table 1—Micro-fracturing and Closure Pressure Analysis Summary

Station	Formation	Hydrostatic Pressure	Breakdown Pressure	ISIP	Propagation Pressure	Closure Pressure
#	-	psi	psi	psi	psi	psi
1	Sand A	1515.8	2209.1	1547.9	1582.6	1536.6
2	Sand B	1560.0	2593.8	1820.1	1859.6	1759.7
4	Sand C	1627.0	2915.8	2023.6	2108.5	2010.6

Station # 1 in Sand A.

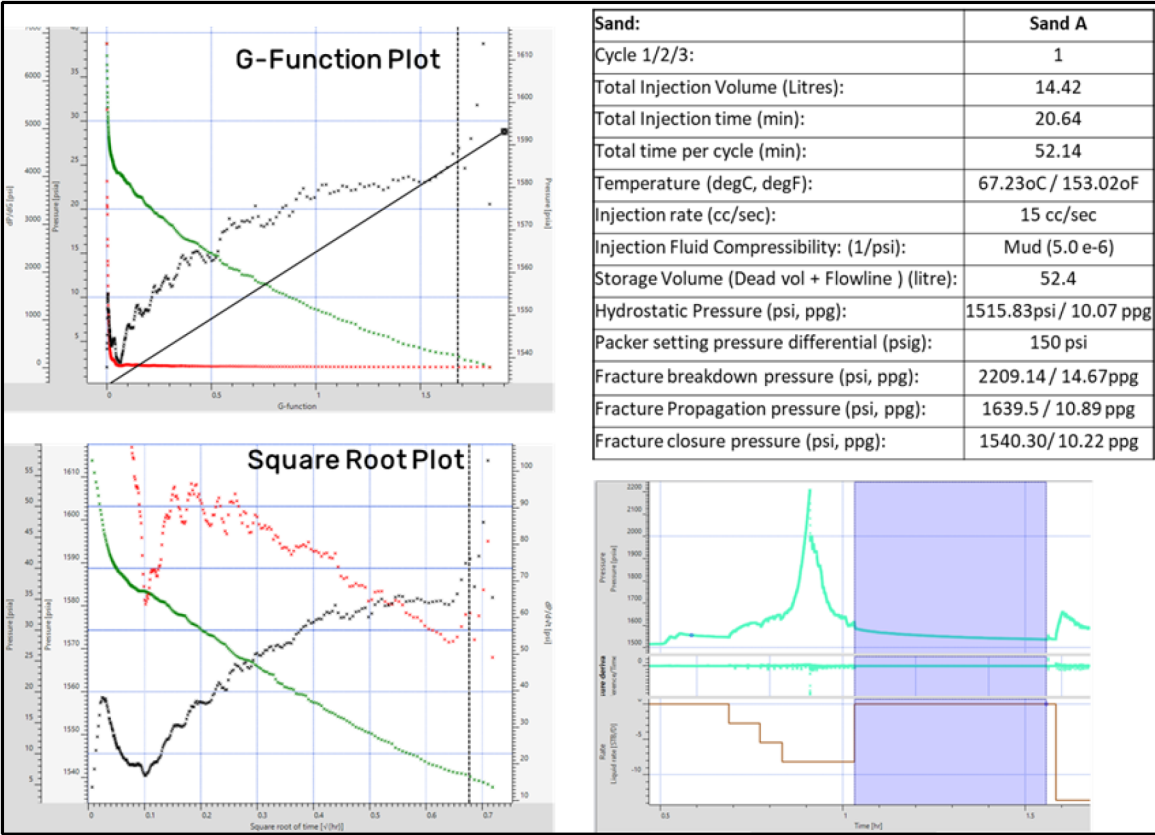


Figure 7—Natural fall-off Test Interpretation - Station #1 Cycle #1

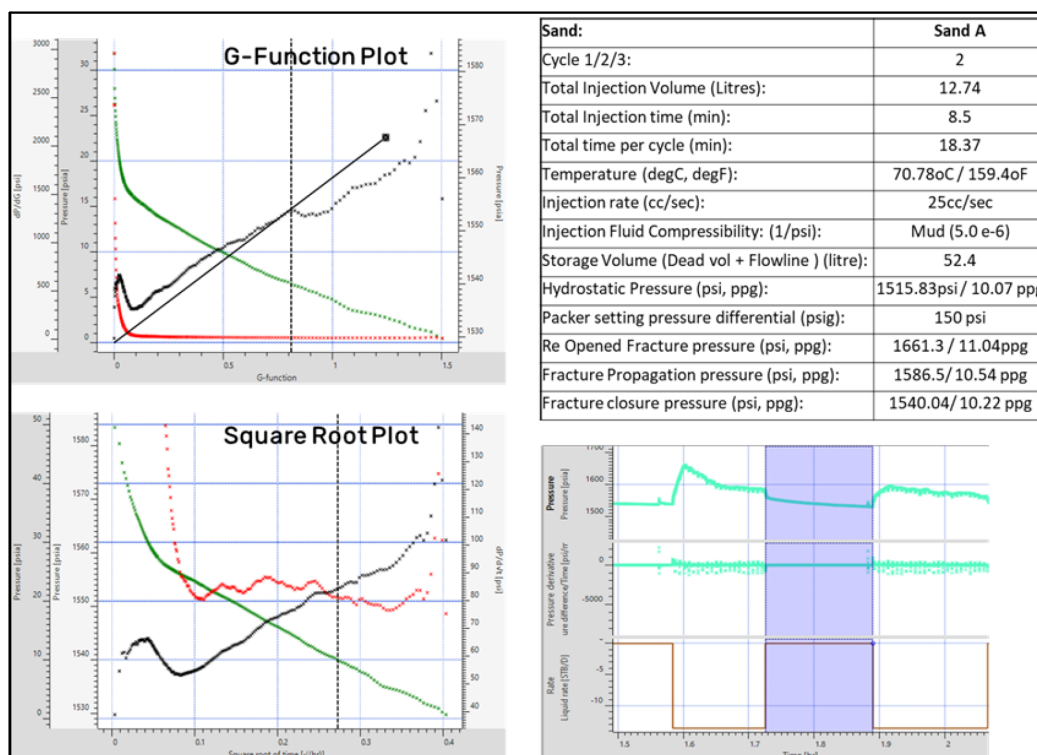


Figure 8—Natural fall-off Test Interpretation - Station #1 Cycle #2

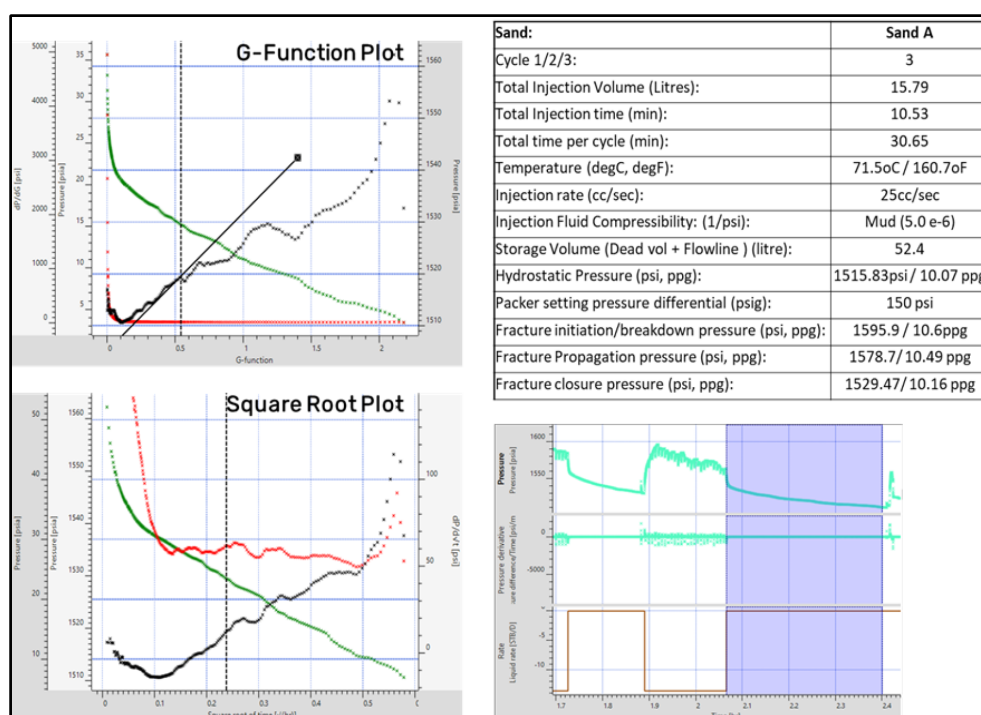


Figure 9—Natural fall-off Test Interpretation - Station #1 Cycle #3

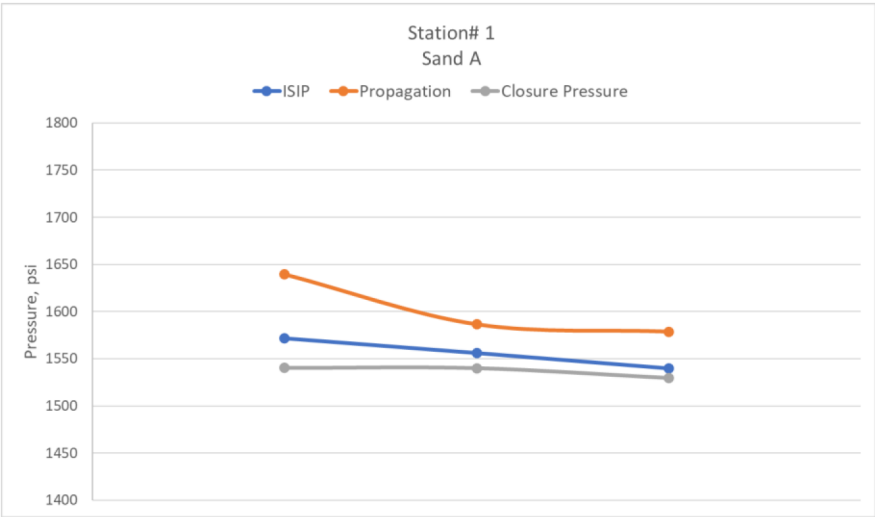


Figure 10—Reconciliation Plot for Station # 1

Station # 2 in Sand B.

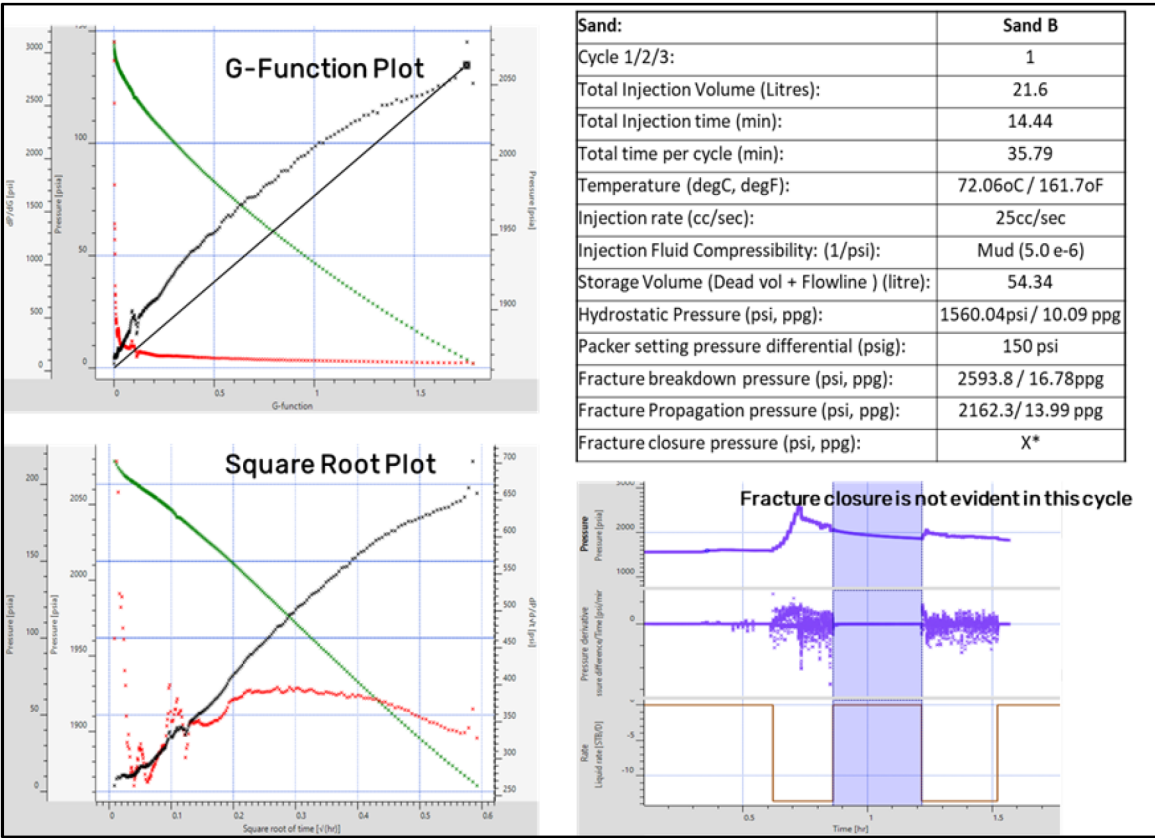


Figure 11—Natural fall-off Test Interpretation - Station #2 Cycle #1

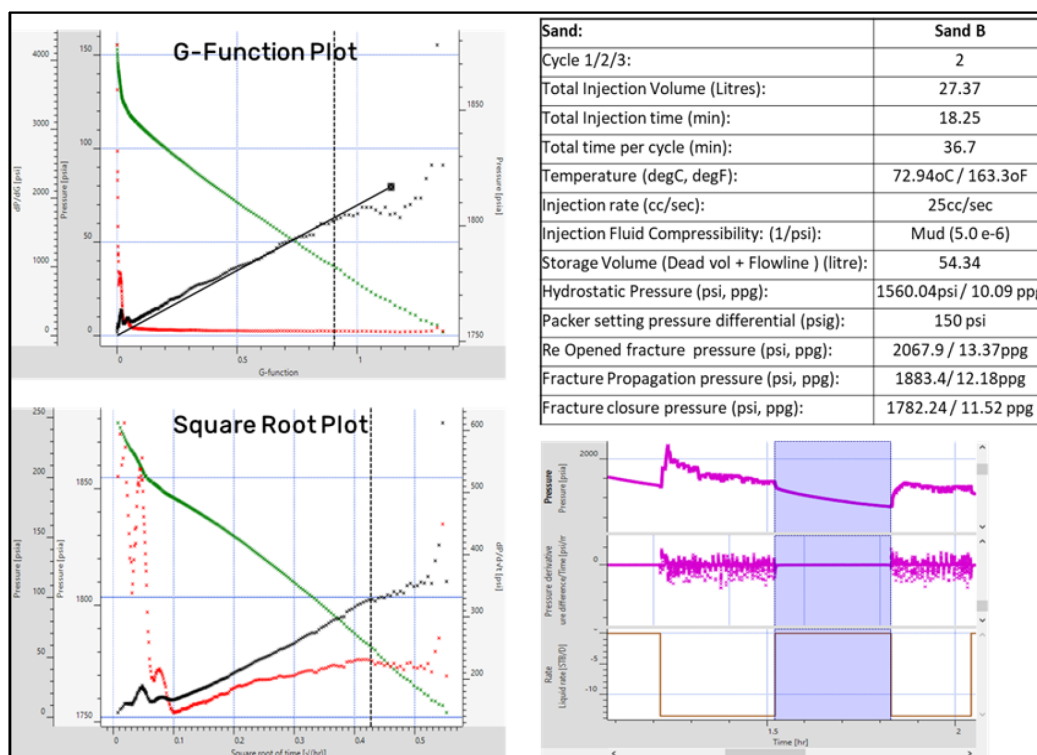


Figure 12—Natural fall-off Test Interpretation - Station #2 Cycle #2

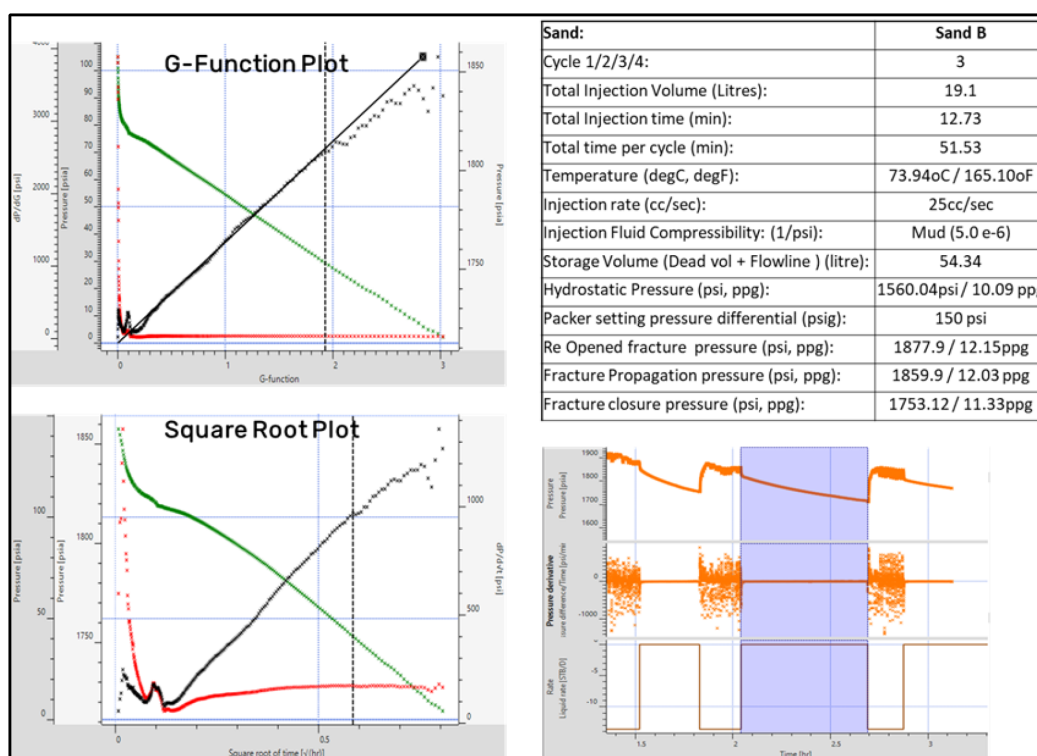


Figure 13—Natural fall-off Test Interpretation - Station #2 Cycle #3

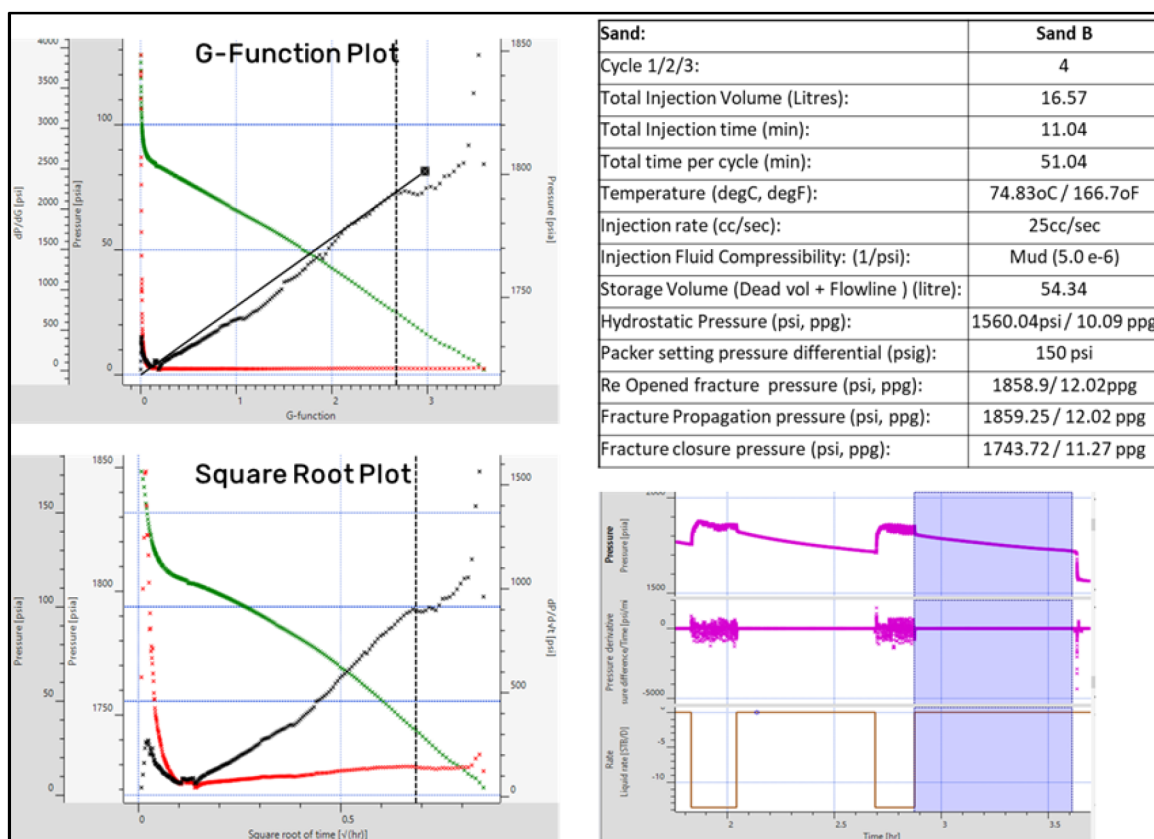


Figure 14—Natural fall-off Test Interpretation - Station #2 Cycle #4

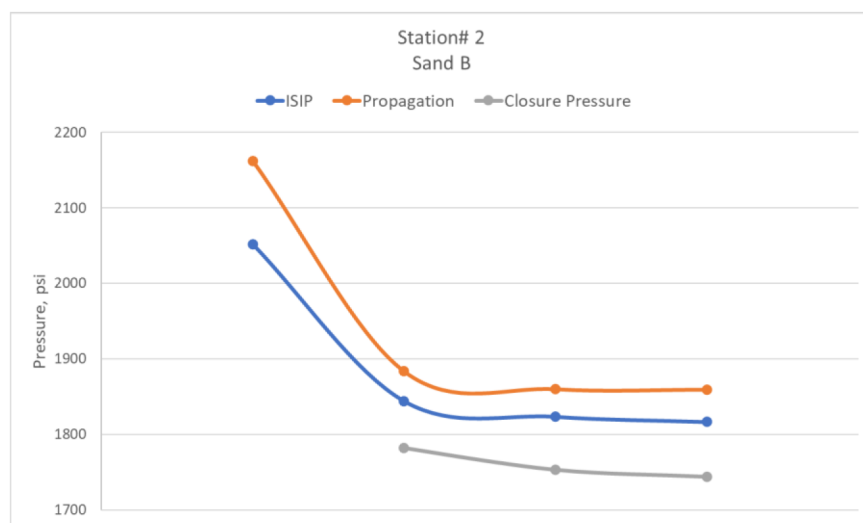


Figure 15—Reconciliation Plot for Station # 2

Station # 4 in Sand C. All microfracture cycles at the aforementioned stations were systematically analyzed to determine the Instantaneous Shut-In Pressure (ISIP) and fracture closure pressure. In each case, the first cycle consistently exhibited a slightly elevated closure pressure relative to subsequent cycles. However, the closure pressures derived from the later cycles were highly repeatable, indicating reliable measurement consistency. This behavior is clearly illustrated in the Reconciliation plots presented in Figs. 10, 15, and 19, which overlay pressure data from all injection cycles to facilitate a rapid visual assessment of repeatability and data quality.

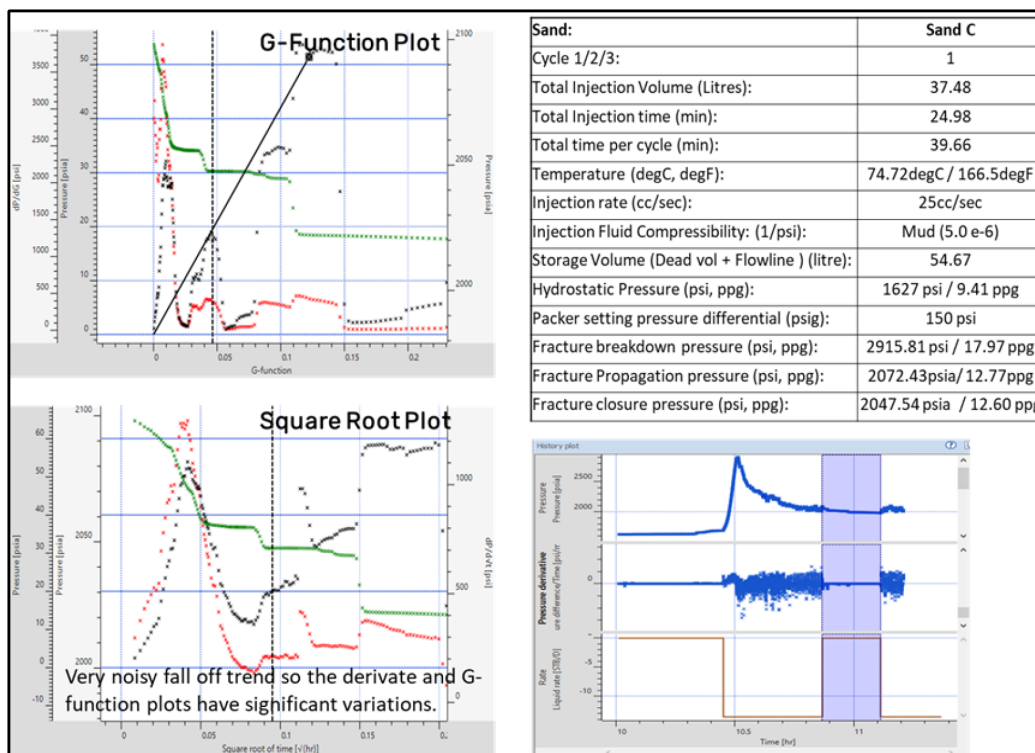


Figure 16—Natural fall-off Test Interpretation - Station #4 Cycle #1

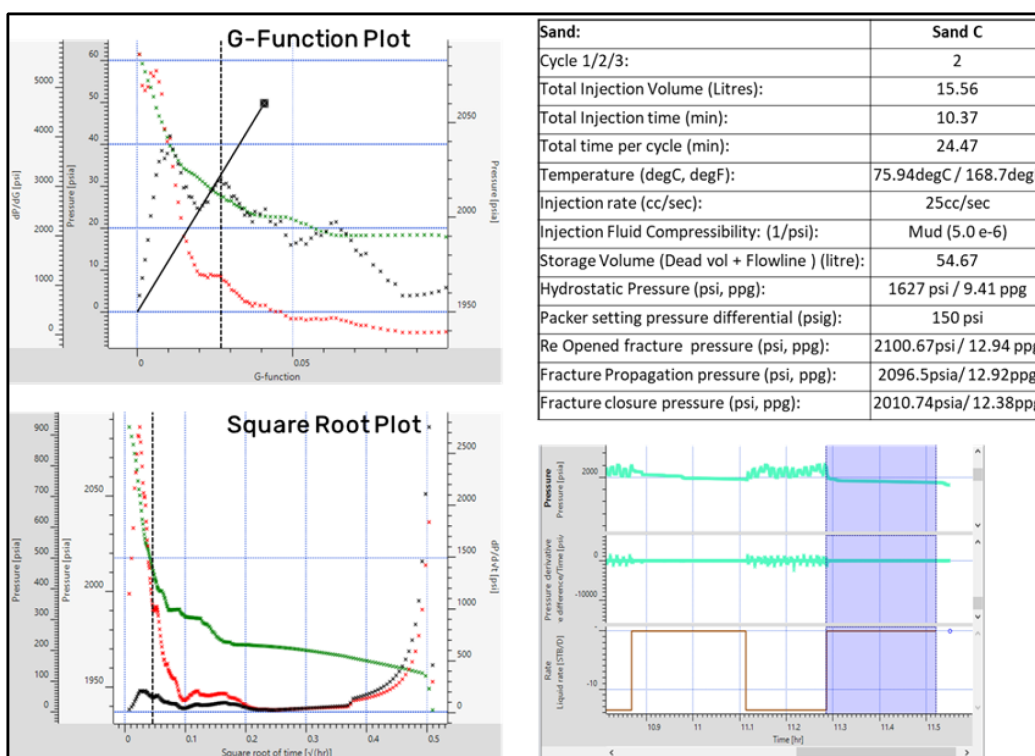


Figure 17—Natural fall-off Test Interpretation - Station #4 Cycle #2

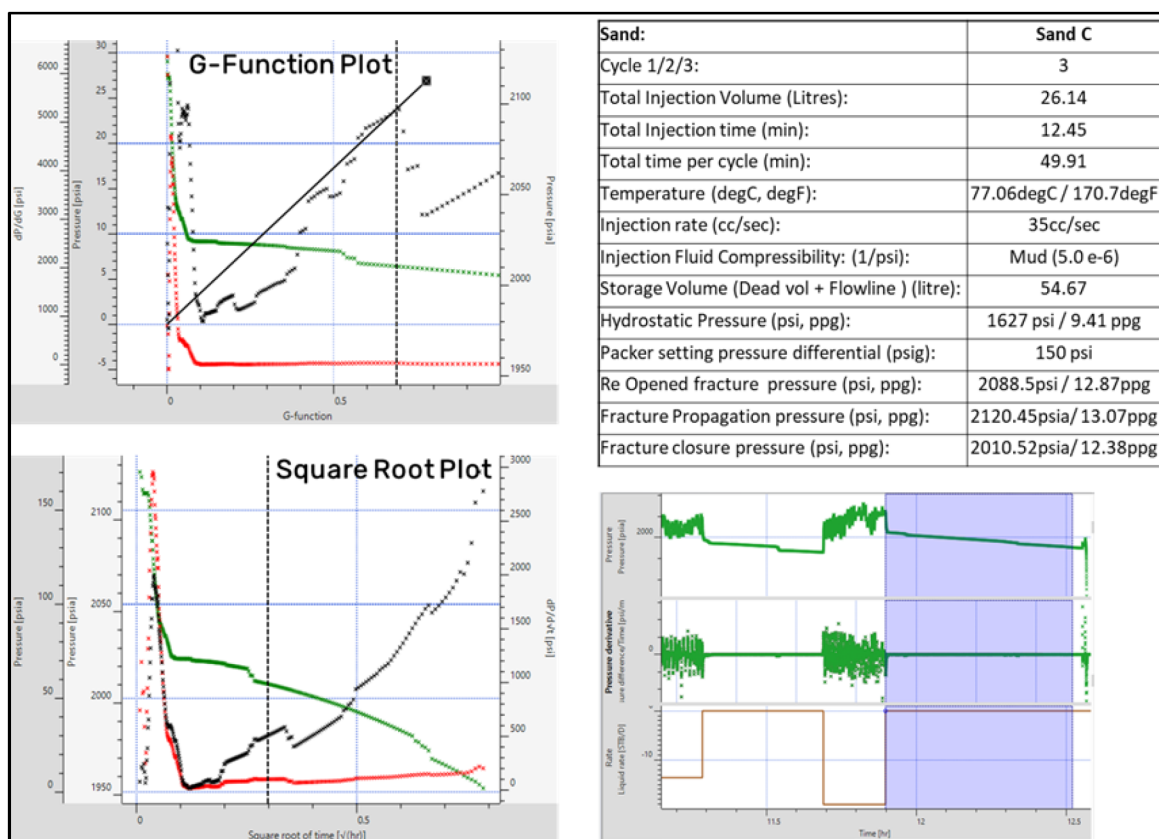


Figure 18—Natural fall-off Test Interpretation - Station #4 Cycle #3

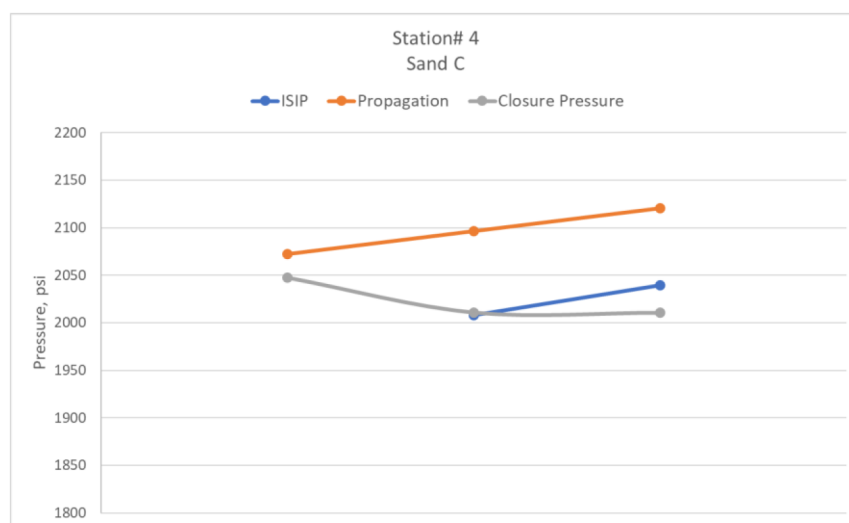


Figure 19—Reconciliation Plot for Station #4

An additional observation during the G-function analysis was the presence of a discernible deviation, or "hump," occurring above the initial linear trend prior to the expected straight-line behavior. This feature is indicative of pressure-dependent leakoff (PDL), a phenomenon commonly attributed to the activation of secondary fractures intersecting the primary induced fracture. The presence of PDL suggests a more complex fracture network and enhanced fluid loss pathways, which may have implications for both fracture propagation and the estimation of closure stress.

Fig. 20 summarizes closure and breakdown pressures versus mud pressure for all stations, showing that measured values fell within safe margins of the designed mud weight windows.

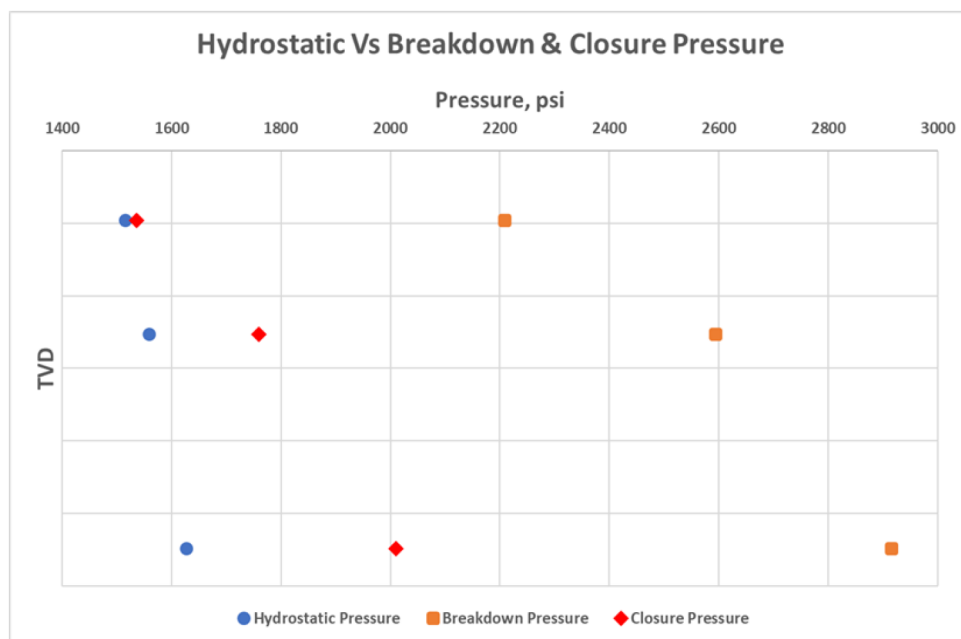


Figure 20—Hydrostatic Versus Microfrac Breakdown & Closure Pressure Profile

Discussion

Micro-fracturing serves as a critical tool in geomechanics by providing direct, in-situ measurements of subsurface stress conditions, particularly the minimum horizontal stress (Sh_{min}). This capability is especially valuable in environments where conventional well log or laboratory-based estimates are compromised due to missing data or altered formation properties, often resulting from reservoir depletion. In such contexts, micro-fracturing offers a reliable, field-based calibration method that enhances geomechanical model fidelity by reducing reliance on assumed rock mechanical properties. The case study discussed here illustrates how iterative integration of microfracture data into the geomechanical workflow leads to progressively refined subsurface models, with each new well contributing to an improved understanding of the stress field. This is particularly important in depleted zones, where significant reductions in pore pressure can lead to lower minimum principal stresses, necessitating frequent updates to the safe mud weight window to mitigate risks such as lost circulation or unintentional fracturing of weakened formations.

The geomechanical model, however, does not remain static and is continuously refined with real-time and post-drill data. Inputs such as sonic logs, pressure-while-drilling (PWD) data, and memory-gauge pressure data offer valuable post-drill insights that improve model accuracy. In this study, the incorporation of formation breakdown and closure pressures from micro-fracturing enabled recalibration of fracture gradients, leading to more informed well trajectory planning and casing design. A noteworthy finding was the 14% deviation between pre-test estimated and measured values of Sh_{min} , which underscores the limitations of log-based stress predictions, particularly in pressure-depleted formations. Such discrepancies are especially critical when operating within narrow drilling margins, where even small errors can pose significant operational risks.

Each microfracture test cycle yields a comprehensive dataset that includes pressure build-up, formation breakdown, fracture propagation, shut-in, and fall-off behavior. The quality and reliability of this data depend heavily on tool calibration, fluid viscosity, controlled injection rates, and effective packer isolation.

In this case, both square root of time and G-function analyses were employed to interpret the fall-off data, offering complementary methods for identifying the Instantaneous Shut-In Pressure (ISIP) and Fracture Closure Pressure (FCP). Observations from the tests revealed a high degree of cycle-to-cycle repeatability at most stations, suggesting minimal influence from near-wellbore damage and well-executed test procedures. In high-permeability zones, however, rapid pressure fall-off posed challenges for accurately capturing closure events, necessitating high-resolution pressure recording. At one location (Station #3), the inability to achieve formation breakdown highlighted either an elevated formation strength or limitations in the pressure capacity of the deployed tool, underscoring the need for higher-rated equipment in future operations under similar conditions.

The practical value of micro-fracturing was most clearly demonstrated through its application in planning two subsequent wells. With recalibrated fracture gradients based on field-acquired data, the drilling team was able to avoid deploying Managed Pressure Drilling (MPD) systems, thereby reducing operational complexity and cost. Additionally, a wellbore strengthening strategy was successfully implemented to maintain hole integrity, even in highly depleted formations. Accurate knowledge of S_{h_min} allowed for confident mud weight management within a narrow margin, significantly lowering the risk of well control incidents. These benefits translated not only to cost avoidance but also to enhanced operational safety and reliability – critical considerations in mature fields with complex depletion histories.

From an economic perspective, although micro-fracturing introduces an initial cost burden associated with tool deployment and test execution, the return on investment is substantial. In this case, the cost savings from avoiding MPD and preventing instability-related non-productive time outweighed the expense of the microfracture program. Furthermore, the refinement of the geomechanical model offers long-term value by supporting better planning for sidetracks, stimulation programs, and enhanced oil recovery (EOR) efforts, all of which depend on an accurate understanding of subsurface stress conditions.

Conclusions

This case study has demonstrated how micro-fracturing, when integrated with geomechanical modeling, serves as a powerful technique to de-risk drilling in depleted and high-permeability reservoirs. The following key conclusions can be drawn:

1. **Model Validation:** Micro-fracturing provides the most direct and reliable method for in-situ stress validation, particularly in reservoirs where core or shear slowness logs are absent.
2. **Operational Optimization:** By integrating real-time and post-drill data, including microfrac-derived stress measurements, operators can optimize mud weight, casing design, and trajectory planning.
3. **Tool and Test Design:** High-quality tools with adequate pressure ratings, along with multiple cycles per test station, enhance the quality of results and reduce uncertainties from borehole effects.
4. **Drilling Efficiency:** The refined geomechanical model enabled two subsequent wells to be drilled without the use of MPD. This translated into lower costs, safer operations, and a high-confidence drilling environment.
5. **Scalability:** The methodology applied here is scalable across similar brownfield projects, particularly in the Middle East and Southeast Asia, where reservoir depletion and high permeability are common.
6. **Recommendations:** Operators should include micro-fracturing as part of the standard workflow for geomechanical modeling in challenging environments. Furthermore, repeated injection/fall-off cycles and robust analytical interpretation are critical to ensuring data reliability.

In sum, this study adds to the growing body of evidence advocating for a tighter integration of real-time field measurements with geomechanical modeling. The micro-fracturing method, though mature, continues to offer new value when employed with rigor and analytical depth, especially in complex reservoir settings.

Acknowledgements

The authors wish to thank Petronas, Halliburton and SLB for allowing publication of this work. They further acknowledge the collaborative efforts of field personnel, service partners, and data analysts who contributed to the successful planning, execution, and interpretation of the micro-fracturing operations described in this study.

References

- Ardila, Mario, Achourov, Vladislav, Gisolf, Adriaan, and Stephen Williams. "Formation Testing, Completion Integrity Evaluation and Geomechanical Applications Using a Wireline Cased-Hole Formation Tester." Paper presented at the SPE Offshore Europe Oil and Gas Conference and Exhibition, Aberdeen, UK, September 2009. doi: <https://doi.org/10.2118/125106-MS>.
- Barree, Robert D., and Jennifer L. Miskimins. "Calculation and Implications of Breakdown Pressures in Directional Wellbore Stimulation." Paper presented at the SPE Hydraulic Fracturing Technology Conference, The Woodlands, Texas, USA, February 2015. doi: <https://doi.org/10.2118/SPE-173356-MS>.
- Bautmans, P., Helstrup, O.A., Constable, K.C., and T. Pfeiffer. "Wireline Stress Testing In Reservoir Sands: Some Geomechanical Considerations and a Case from the Norwegian Continental Shelf" Paper presented at the 47th U.S. Rock Mechanics/Geomechanics Symposium, San Francisco, California, June 2013.
- Fjaer, E., Holt, R.M., Horsrud, P., Raaen, A.M., & Risnes, R. (2008). *Petroleum Related Rock Mechanics* (2nd ed.). Elsevier.
- Franquet, J., Gaz, I., Ibragimov, F., Seryakov, A., and J. Huthwaite. "Caprock Integrity Testing Through Openhole Micro-Fracturing In Caspian Sea Sub-Salt Formations." Paper presented at the Offshore Mediterranean Conference and Exhibition, Ravenna, Italy, March 2011.
- Grieser, W.V. and Bray, J.M. 2007. Identification of Production Potential in Unconventional Reservoirs. Society of Petroleum Engineers. Presented at the Production and Operations Symposium, Oklahoma City, Oklahoma, 31 March-3 April. SPE-106623-MS. <https://doi.org/10.2118/106623-MS>.
- Hamid, S., Jackson, B.M., and J.E. Legenza. "Injection System for Microfrac or Step-Rate Testing." Paper presented at the SPE Eastern Regional Meeting, Columbus, Ohio, October 1990. doi: <https://doi.org/10.2118/21267-MS>
- Ishijima, Y. and Roegiers, J.C. 1983. Fracture Initiation And Breakdown Pressure -Are They Similar? Presented at the 24th U.S. Symposium on Rock Mechanics (USRMS), College Station, Texas, USA, 20-23 June. ARMA-83-0761.
- Ihab, Tarek, Naial, Radwan, Moronkeji, Dee A., Franquet, Javier A., and Steven S. Smith. "Wireline Straddle Packer Microfrac Testing Enables Tectonic Lateral Strain Calibration in Carbonate Reservoirs." Paper presented at the International Petroleum Technology Conference, Doha, Qatar, January 2014. doi: <https://doi.org/10.2523/IPTC-17301-MS>
- Kuhlman, R.D., Microfrac tests optimize frac jobs. *Oil & Gas Journal*, Jan 22 1990
- Lecampion, B., Abbas, S., and Prioul, R. 2013. Competition Between Transverse and Axial Hydraulic Fractures In Horizontal Wells. Presented at the SPE Hydraulic Fracturing Technology Conference, The Woodlands, Texas, USA. 4-6 February. <https://doi.org/10.2118/163848-MS>.
- Malik, Mayank, Singh, Amit, and Manish K. Lal. "Stress Profiling with Microfracturing and Sonic Logs in Antelope Shale." Paper presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA, September 2015. doi: <https://doi.org/10.2118/175002-MS>.
- Mishra, Vinay K., Lywood, Peter, and Cosan Ayan. "Application of Wireline Stress Testing for SAGD Caprock Integrity." Paper presented at the Canadian Unconventional Resources Conference, Calgary, Alberta, Canada, November 2011. doi: <https://doi.org/10.2118/149456-MS>.
- Warpinski, N.R., Branagan, P., Wilmer, R., 1985. In-situ stress measurements at U.S. DOE's multiwall experiment site, *Mesaverde Group*, Rifle, Colorado. SPE 12142, JPT, 527-536.
- Zoback, M.D. (2007). *Reservoir Geomechanics*. Cambridge University Press.